Generating Synthetic Face Images with Deep Convolutional Generative Adversarial Networks

Maurizio Pinto

Woolf University

Generative AI Applications

Udacity

May 1, 2026

Generating Synthetic Face Images with Deep Convolutional Generative Adversarial Networks

Overview

This project addresses the task of synthesizing realistic human face images using a Deep Convolutional Generative Adversarial Network (DCGAN). The DCGAN architecture (Radford et al., 2015) implements the generative adversarial framework (Goodfellow et al., 2014), in which two neural networks — a Generator that produces images and a Discriminator that evaluates them — are trained in opposition. The model is trained on a subset of 50,000 images from the CelebA dataset (Liu et al., 2015), a large-scale collection of aligned celebrity face photographs. The goal is to produce 64×64 face images that are visually comparable to real photographs.

Dataset or Prompt Description

The CelebA dataset (Liu et al., 2015) contains 202,599 aligned and cropped face images of celebrities, originally captured at 178×218 pixel resolution in RGB format. The images feature a wide range of facial attributes including varied expressions, hair styles, accessories, and lighting conditions, though they are all photographed against relatively uniform backgrounds.

For this project, a subset of 50,000 images was selected for computational efficiency while still providing sufficient diversity for the Generator to learn meaningful face representations. All images were resized to 64×64 pixels and normalized to the range [-1, 1] to match the Generator's Tanh output activation. The 64×64 resolution follows the standard DCGAN configuration from Radford et al. (2015) and is small enough to enable stable training on a single GPU while still producing recognizable face images.

An important characteristic of this dataset is its demographic composition. Because CelebA consists of celebrity photographs, it is not representative of the general population — it disproportionately features young, conventionally attractive individuals with a skew toward lighter skin tones. This bias is directly relevant to the ethical considerations discussed later in this report.

Model Design and Training Approach

The DCGAN consists of two networks with mirrored architectures. The Generator takes a 100-dimensional noise vector sampled from a standard normal distribution and progressively upsamples it through five transposed convolutional layers to produce a 64×64×3 image. The channel progression is 512→256→128→64→3, with each layer doubling the spatial resolution while halving the number of feature channels. Batch normalization is applied in all layers except the final output layer, and ReLU activations are used throughout, with a Tanh activation on the output to produce values in [-1, 1]. The Generator contains approximately 3.6 million parameters.

The Discriminator performs the inverse operation, taking a 64×64×3 image and progressively downsampling through five strided convolutional layers to produce a single probability value. The channel progression is 64→128→256→512→1, with batch normalization in all layers except the first and LeakyReLU(0.2) activations that prevent sparse gradients. The Discriminator contains approximately 2.8 million parameters.

Key architectural choices follow the DCGAN guidelines established by Radford et al. (2015): strided convolutions replace pooling layers to allow the networks to learn their own spatial aggregation, batch normalization stabilizes training by normalizing activations, and all weights are initialized from a normal distribution with mean 0 and standard deviation 0.02. The use of LeakyReLU in the Discriminator rather than standard ReLU ensures that gradient information flows even for negative inputs, which is critical for training the Generator through the adversarial signal.

The model is trained using binary cross-entropy loss, which is the standard loss function for the original GAN formulation (Goodfellow et al., 2014). Both networks use the Adam optimizer with a learning rate of 0.0002 and a first momentum coefficient (beta1) of 0.5, as recommended by Radford et al. (2015) — the lower momentum compared to the default of 0.9 reduces training oscillation. The batch size is 128, and the model is trained for 50 epochs, yielding approximately 19,500 training iterations.

Output Evaluation and Interpretation

After 50 epochs of training, the DCGAN generates face images that demonstrate plausible overall structures. The generated faces exhibit correct facial proportions with eyes, nose, and mouth positioned appropriately within oval-shaped head outlines. The model captures a variety of attributes including different hair styles, skin tones, head orientations, and expressions, indicating that the latent space is well-structured and not collapsed to a single mode.

The training progression visualization shows clear improvement over time. Epoch 1 produces images that are predominantly noise with faint face-like structures, while by epoch 25 the faces are clearly recognizable. The loss curves show moderate oscillation between the Generator and Discriminator losses throughout training, which is characteristic of healthy adversarial training — neither network has completely dominated the other, and both continue to improve.

However, several limitations are evident in the generated outputs. Fine details such as individual eyes, teeth, and hair strand boundaries appear blurry or slightly distorted. This is a well-known limitation of DCGANs operating at 64×64 resolution — the network lacks the capacity to render high-frequency details with precision. A specific failure case observed in multiple samples involves asymmetric facial features, where one eye or one side of the mouth appears differently positioned than the other. In several generated faces, the region around the ears and jawline contains artifacts that resemble blending between different face templates. These failure cases suggest that while the Generator has learned the overall distribution of face structures, it has not fully converged in all regions of the latent space.

The latent space interpolation between two randomly generated faces produces smooth, continuous transitions, with intermediate faces maintaining plausible features throughout. This indicates that the Generator has learned a meaningful representation rather than memorizing isolated examples from the training data.

Ethical Considerations and Responsible Use

Generative models that produce realistic face images raise significant ethical concerns, particularly regarding the creation of deepfakes — synthetic media designed to deceive viewers into believing fabricated content is real (Chesney & Citron, 2019). The DCGAN implemented in this project, while limited to 64×64 resolution, demonstrates that even a relatively simple architecture can produce faces that appear genuine at a glance. As generative models improve in resolution and quality, the potential for misuse in creating fraudulent identity documents, fake social media profiles, or non-consensual synthetic imagery becomes a serious concern.

A specific ethical issue relevant to this project is the demographic bias embedded in the CelebA training data. Because the dataset consists of celebrity photographs, it overrepresents young, conventionally attractive individuals with lighter skin tones. The generated faces reflect this bias — the model is more likely to produce faces that match the overrepresented demographics and may struggle to generate realistic faces of underrepresented groups. In a real-world application, such as an AI avatar generator or a data augmentation tool, this bias would systematically exclude or poorly represent certain populations.

Additionally, the individuals whose photographs appear in CelebA did not consent to having their likenesses used to train a generative model. While the images were collected from publicly accessible sources, the question of whether public availability constitutes consent for synthetic face generation remains an open ethical debate. Responsible development of generative face models requires transparent data sourcing, diverse training data, clear disclosure that images are AI-generated, and safeguards against deployment in contexts where deception could cause harm.

Limitations and Future Improvements

Several limitations should be acknowledged. The 64×64 output resolution is too low for most practical applications, and the DCGAN architecture struggles with fine details such as individual facial features and texture patterns. The absence of a quantitative evaluation metric such as the Frechet Inception Distance (FID) makes it difficult to objectively compare the model's output quality against other approaches or track improvement over training. Training stability, while acceptable in this run, is not guaranteed — GANs are known for their sensitivity to hyperparameters and random initialization, and different runs may produce significantly different results.

Future improvements could address these limitations through several approaches. Progressive Growing of GANs (Karras et al., 2018) would allow the model to incrementally increase output resolution from 4×4 to 128×128 or higher, producing significantly more detailed faces. Spectral normalization of the Discriminator's weight matrices would constrain the

Lipschitz continuity of the network, leading to more stable training. Using the full CelebA dataset (200K images) rather than the 50K subset would provide greater diversity in the training distribution. Finally, a conditional GAN architecture would enable control over specific attributes such as hair color, gender, or expression, making the model more useful for targeted generation tasks.

References

Chesney, R., & Citron, D. (2019). Deep fakes: A looming challenge for privacy, democracy, and national security. California Law Review, 107(6), 1753–1820.

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. Advances in Neural Information Processing Systems, 27.

Liu, Z., Luo, P., Wang, X., & Tang, X. (2015). Deep learning face attributes in the wild. Proceedings of the IEEE International Conference on Computer Vision, 3730–3738.

Radford, A., Metz, L., & Chintala, S. (2015). Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434.

Karras, T., Aila, T., Laine, S., & Lehtinen, J. (2018). Progressive growing of GANs for improved quality, stability, and variation. Proceedings of the International Conference on Learning Representations.